Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Computer Application

MCA | Sem-3 | Winter-2025

Course Code	:	2305CS303	**Date**	:	11-10-2025
Course Name	:	Python for Data Science	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	List any four advantages of Python over other programming languages.	**4**
	(B)	What is the difference between identity operators and membership operators in Python? Explain with small examples.	**3**
		OR	
		What are magic commands in Python? Give any two examples commonly used in Jupyter Notebook.	
	(C)	What is Datatype? List out different Datatypes in Python with example.	**7**
		OR	
		Explain with examples how the break, continue, and pass keywords affect the flow of loops in Python.	
Q.2	**(A)**	List any four commonly used string methods in Python and explain their purpose.	**4**
	(B)	What is a docstring in Python? How is it useful? Provide a small example.	**3**
		OR	
		Explain lambda function in Python. Write a simple lambda function to double a number.	
	(C)	Compare List, Tuple, Set, and Dictionary in Python.	**7**
		OR	
		Explain the different types of function arguments in Python with examples.	
Q.3	**(A)**	Explain any four attributes of a NumPy array with suitable examples.	**4**
	(B)	What are array creation routines in NumPy? Explain any three with examples.	**3**
		OR	
		Write short note on conditional selection in NumPy with an example.	
	(C)	Write a Python program using Beautiful Soup to extract all hyperlinks (<a> tags with href attributes) from an HTML document. Explain the steps involved.	**7**
		OR	

Consider the following NumPy array:

```
import numpy as np
arr = np.array([[5, 10, 15, 20],
                [25, 30, 35, 40],
                [45, 50, 55, 60],
                [65, 70, 75, 80]])
```

Using indexing and slicing, write Python code to perform the following operations:
a) Extract the first two rows of the array.
b) Extract the last column of all rows.
c) Extract the subarray [[30, 35], [50, 55]].
d) Replace all elements in the second row with 0.

Q.4 **(A)** Explain how Pandas handles missing data. Discuss dropna(), fillna(), and interpolate() with examples. **4**

(B) Write a short note on setting and resetting index in a Pandas DataFrame. **3**

OR

What is a Time Series in Pandas? Give an example of creating a simple time series.

(C) Consider the following DataFrame created from a dictionary: **7**

```
import pandas as pd
data = {
   'EmpID': [101, 102, 103, 104, 105],
   'Name': ['Aarav', 'Diya', 'Karan', 'Meera', 'Sahil'],
   'Dept': ['HR', 'IT', 'IT', 'Finance', 'HR'],
   'Salary': [40000, 55000, 60000, 45000, 70000],
   'Bonus': [5000, None, 8000, None, 10000]
}
df = pd.DataFrame(data)
```

Answer the following questions using Pandas operations:
a) Display the names of all employees working in the IT department.
b) Find the employees having Salary greater than 50,000.
c) Replace all missing values in the Bonus column with 0.
d) Group the employees by Department and find the average salary for each department.

OR

Consider the following DataFrame created in Pandas:

```
import pandas as pd
data = {
   'RollNo': [1, 2, 3, 4, 5],
   'Name': ['Aarav', 'Diya', 'Karan', 'Meera', 'Sahil'],
   'Math': [78, 85, 92, 67, 80],
   'Science': [88, 79, 95, 70, 85]
}
df = pd.DataFrame(data)
```

Answer the following using Pandas:
a) Display the names of all students who scored more than 80 in Math.
b) Display only the RollNo and Name columns.
c) Add a new column Total which is the sum of Math and Science marks.
d) Find the average marks in Science.

Q.5 **(A)** What are gridlines and ticks in a plot? Write commands to enable gridlines and customize tick marks in a Matplotlib chart. **4**

(B) Write Python code to plot a boxplot for the dataset: **3**
scores = [55, 67, 89, 45, 78, 90, 100, 60]
Label the y-axis as "Marks" and give the plot a title "Student Scores".

OR

Write Python code to plot a histogram for the following dataset using 5 bins:
Ages = [18, 22, 25, 19, 30, 27, 24, 21, 20, 23, 26]
Include axis labels and title.

(C) Write a Python program to plot the marks of 5 students in two subjects (Math and Science) on the same line chart using matplotlib with different line styles, colors, and markers. Add axis labels, title, legend, and annotation to highlight the highest mark. Export the graph to a PDF file. **7**

OR

Write a Python program to plot a bar chart using matplotlib showing sales of 5 products. Display the value of each bar on top, add axis labels, title, and custom colors for each bar. Export the chart as a PNG image.

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